

Akshay Borse

425-336-9852 | amsborse@gmail.com | [Linkedin](#) | [Github](#) | [Portfolio](#) | Auburn, WA

PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years building and scaling large-scale platforms, distributed systems, and AI-powered products. Experience spans product development, platform engineering, architecture modernization, and applied AI, with systems operating at 100K+ TPS, platforms adopted across 60+ services, and products delivering \$28M+ in measured annualized impact.

EXPERIENCE

Senior Software Engineer | Microsoft | Redmond, WA

Sep 2025 to Aug 2026

- **Built the historical risk foundation for AI Agent Risk Detection**, reconstructing 90 days of activity across 40+ environments to eliminate onboarding blind spots and enable day-one risk evaluation for 237K+ agents across 13K organizations.
- **Drove architecture across Purview IRM, Entra, and DLP** for Microsoft's AI-agent Adaptive Protection pipeline, owning design, implementation, and launch from risk scoring through Conditional Access enforcement and shipped the capability with Microsoft 365 E7.
- **Led the redesign of AI-agent ingestion for independent scale and fault isolation**, separating agent and human workloads across 28 deployments processing 135K+ security signals/day and protecting existing Insider Risk Management pipelines as agent traffic grew.
- **Led AI Risk Detection's sovereign-cloud architecture and rollout across GCC, GCCH and DoD**, engineering deployment and disaster-recovery infrastructure across 53 resource groups and 51 deployments and establishing the first Gov-cloud Kusto monitoring infrastructure.
- **Connected investigations across Microsoft's security ecosystem**, linking related risk across Entra, Defender, Sentinel, Microsoft Graph, and Insider Risk Management and increasing customer engagement with related cross-product investigations 16%.
- **Designed an AI-assisted testing workflow** using agent skills and Playwright to provision test tenants, configure Copilot Studio agents with tools/policies, trigger controlled security risks and alerts, and validate end-to-end behavior.
- **Designed an AI-assisted debugging workflow spanning three repositories**, combining historical incident analysis with Playwright to reproduce issues, implement and validate fixes, capture evidence, and create pull requests.

Software Engineer 2 | Amazon | Seattle, WA

Aug 2019 – Aug 2025

- **Built a 0→1 developer platform for 60+ microservices**, creating a one-click CDK-based workflow that standardized infrastructure, authentication, observability, dependencies, CI/CD, and deployment into a production-ready service; cut setup from 2 weeks to under 1 hour, consolidated 9 pipelines, and saved 40–50 engineering weeks across ~40 engineers and 3 teams.
- **Led a monolith-to-microservices transformation at 100K+ TPS**, building and launching 15+ services and establishing shared REST/gRPC orchestration supporting 10M+ daily transactions while maintaining behavioral parity throughout migration.
- **Designed and owned the service powering ML-predicted delivery ranges**, integrating a quantile-regression model and coordinating changes across inventory planning and fulfillment optimization systems to replace single-date promises; delivered \$28M in annualized impact.
- **Replaced static delivery-risk rules with ML-driven decisioning**, using XGBoost on SageMaker to identify at-risk orders and trigger recovery, customer notifications, and alternate-seller recommendations, reducing cancellations and contributing to a 7% increase in sales.
- **Launched inventory decisioning systems processing 200K+ events/day**, launching Just-In-Stock and Restock Alerting with 30-day order replay and real-time inventory signals; increased restocked-item sales 13% and early deliveries 16%.
- **Architected built a RAG-based product recommendation system for out-of-stock items**, to surface relevant alternatives grounded in real-time inventory and delivery constraints, keeping availability and fulfillment validation deterministic. Increased eligible product sales by 23%.
- **Led design and cross-functional adoption of an LLM-powered localization platform**, using Bedrock with vocabulary-constrained generation and deterministic validation to personalize 37 email templates across 28 locales, with pre-approved template fallbacks on validation failure to address hallucination and compliance risks.
- **Owned the annual operational improvement roadmap across 60+ services**, driving planning and execution across 15+ engineers and reducing Sev2 incidents 40% in two months; led readiness across 90 services for Prime Day and Black Friday with zero major incidents.
- **Turned synchronized delivery for large Amazon Business orders into a first-class fulfillment capability**, driving changes across planning, inventory, and fulfillment systems to honor coordinated delivery dates; replaced manual engineering workflows with self-service tooling, eliminated 100–300 engineering hours/year, and improved GCCP 11%.
- **Standardized observability and container infrastructure across 60+ services**, introducing shared logging and Docker libraries that cut CloudWatch spend \$250K+/month, container size 83%, and build time 50%, with container standards adopted across teams.

Software Engineering Intern | Liquiron | San Jose, CA

Dec 2018 to Jan 2019

- **Migrated 2 backend services to serverless**, modernizing authentication with OAuth 2.0/Passport.js and Firebase.
- **Built the API layer for a single-page application**, delivering 4 REST APIs, 10 frontend capabilities and reducing server load by 36%.

Software Engineer | Persistent Systems | Pune, India

Sep 2016 to Jul 2017

- **Built a real-time CNN classification system for images and video**, achieving 87% accuracy across 7 sentiment categories.
- **Modernized legacy web applications into a single-page architecture**, improving response time 25% and unreliable-network performance 9%; introduced an engineering feedback system adopted by 6 teams.

Software Engineering Intern | Soft Corner | Pune, India

Jun 2015 to Jun 2016

- **Improved F2FS garbage-collection efficiency 15% through Linux kernel-level performance engineering**, instrumenting filesystem behavior and benchmarking cost-benefit and greedy cleaning policies to optimize latency.

EDUCATION

- Santa Clara University, Santa Clara, CA | Master's in Computer Science and Engineering. Sep 2017 to Jun 2019
- Pune Institute of Computer Technology (PICT), India | Bachelor's in Computer Science and Engineering. Jun 2012 to Jun 2016

FEATURED PROJECTS

- **LeetDesign**: Executable system-design simulator for modeling scalability, fault tolerance, workloads, and architectural trade-offs.
- **FinanceOS**: AI Financial Intelligence Platform: Designed a hybrid deterministic + LLM architecture that computes financial facts through deterministic pipelines and uses LLM reasoning only for explanation and recommendations, keeping probabilistic models outside correctness-critical calculations.

SKILLS

- **Languages**: Java • C# • Python • C++ • C • TypeScript • JavaScript • SQL
- **Architecture & Systems**: Distributed Systems • System Design • Microservices • Event-Driven Architecture • REST APIs • gRPC • Concurrency • Fault Tolerance • High Availability • Linux
- **Cloud & Platform**: AWS • Azure • Platform Engineering • Developer Platforms • Infrastructure as Code (IaC) • CI/CD • Docker • ECS • Lambda • Step Functions • DynamoDB • Cosmos DB • Observability • SLOs
- **AI/ML & Security**: Generative AI • LLMs • RAG • AI Agents • AI Infrastructure • AI Governance • Machine Learning • NLP • XGBoost • Bedrock • SageMaker • AI Security • DLP • Entra • Purview • Playwright

[HackerRank](#) | [Publication](#) | [Medium](#) | [Substack](#) | [X](#)